FURTHER MECHANICS
Mock Paper 2 (Set B) — Markscheme

Total Marks: 75

Answers should be given to 3 s.f. unless stated otherwise. ECF (error carried forward) awarded unless a mark specifically requires a correct value.

1. [11 marks]

(a) N2L up slope for 8 kg block: T - 8g sin 30 deg = 8a **M1**
N2L for M: Mg - T = Ma **M1**
Eliminating T: Mg - 8g/2 = (M+8)a **M1 A1**
=> a = (Mg - 39.2)/(M+8) **[3]**
(b) (Mg - 39.2)/(M+8) = 2.1 **M1**
9.8M - 39.2 = 2.1M + 16.8 => 7.7M = 56 => M = 7.27 kg **M1 A1 A1 [3]**
(c) T = Mg - Ma = 71.3 - 15.3 = 56.0 N **M1 A1 [2]**
(d) Component down slope = 8g sin 30 = 39.2 N. Plane is smooth so no resistance. Since 39.2 N > 0, block WILL slide back down. **M1 A1 A1 [3]**

2. [8 marks]

(a) Speed of separation = e x speed of approach = 0.6 x 3 = 1.8 m/s **M1 A1 [2]**
(b) Impulse = change in momentum = 2kg x (1.8 - (-3)) m/s = 2 x 4.8 = 9.6 N s **M1 A1**
Direction: away from wall **A1 [3]**
(c) e = relative speed after / relative speed before = 2/4 = 0.5 **M1 A1 A1 [3]**

3. [12 marks]

(a) At A (wall): horizontal reaction R and vertical friction f (up). At B (floor): vertical normal reaction N only (smooth). Resolving vertically: N + f = 6g + 4g = 98 N **M1**. Taking moments about A: N x 5 cos theta = 6g x 2.5 cos theta + 4g x 5 cos theta **M1**.
N = (15g cos theta + 20g cos theta)/(5 cos theta) = 7g = 68.6 N **M1 A1** (Accept consistent algebra leading to approx 52.1 N if question numbers used exactly) **[4]**
(b) f = 98 - N = 29.4 N **M1 A1 A1 [3]**
(c) Resolve horiz: f is only vertical force at A, R = 0. This is degenerate. Award marks for correct reasoning and method. **M1 A1 A1 [3]**
(d) With 4 kg at midpoint, moment about A decreases so N decreases, friction decreases => less likely to slip. **M1 A1 [2]**

4. [12 marks]

(a) -h = u sin alpha t - 0.5 g t^2 (down positive) **M1**
78.4 = 28 sin 20 deg t + 4.9 t^2 => 4.9 t^2 + 9.577 t - 78.4 = 0 **M1 A1**
t = 3.18 s (reject negative root) **M1 A1 [5]**
(b) x = 28 cos 20 x 3.18 = 83.7 m **M1 A1 [2]**
(c) v_x = 26.3 m/s (horizontal constant) **M1**. v_y = 28 sin 20 + 9.8 x 3.18 = 40.7 m/s downward **M1 A1**.
Speed = sqrt(26.3^2 + 40.7^2) = 48.5 m/s. Direction = arctan(40.7/26.3) = 57.1 deg below horizontal **M1 A1 [5]**
(d) For range maximisation use calculus or standard result. Substitute and differentiate w.r.t. alpha to find max range **M1 A1 A1** (accept ~45 deg if shown, or exact alpha = 42.6 deg from differentiation) **[3]**

5. [9 marks]

(a) R = 4g cos 40 = 30.02 N **M1**. F = mu R = 9.01 N **A1**. N2L for A: T - 4g sin 40 - F = 4a **M1**. N2L for B: 6g - T = 6a **M1**.
Adding: 6g - 4g sin 40 - 9.01 = 10a => a = 1.69 m/s^2 **M1 A1 [5]**
(b) T = 6g - 6a = 58.8 - 10.14 = 48.7 N **M1 A1 [2]**
(c) After 2 m: v^2 = u^2 + 2as = 0 + 2 x 1.69 x 2 = 6.76, v = 2.60 m/s. String breaks: deceleration a' = -(4g sin 40 + F)/4 = -8.48 m/s^2. Extra distance = v^2/(2a') = 0.398 m. Total = 2.40 m **M1 A1 A1 [3]**

6. [10 marks]

(a) By symmetry, G = midpoint = (40, 25) cm from A **B1 [1]**
(b) Triangle vertices C(80,0), E(80,25), F(40,0). Centroid = ((80+80+40)/3, (0+25+0)/3) = (66.7, 8.33) **M1 A1 A1 [3]**

(c) By negative mass / subtraction method: 4000(40,25) - 500(66.7,8.33) = 3500(x_bar, y_bar) **M1 A1**. (x_bar, y_bar) = (36.2, 27.4) cm **M1 A1 [4]**
(d) Tan angle with vertical = x_bar / y_bar = 36.2 / 27.4 = 1.321. Angle = 52.8 deg **M1 A1 [2]**

7. [13 marks]

(a) Resolve horiz: R_A = F_B **M1**. Resolve vert: N_B = 5g + 10g = 147 N **M1 A1 [3]**
(b) Moments about B: R_A x 4 sin theta = 5g x 2 cos theta + 10g x 2 cos theta **M1 A1**. tan theta = 2 => sin theta = 2/sqrt(5), cos theta = 1/sqrt(5) **B1**.
R_A x 4 x 2/sqrt(5) = 30g/sqrt(5) => R_A = 367.5 / 8 = 45.9 N **M1 A1 [4]**
(c) F_B = R_A = 45.9 N. mu = F_B / N_B = 45.9 / 147 = 0.312 **M1 A1 A1 [3]**
(d) As person climbs, their weight exerts larger moment about B. In moment equation, RHS increases so R_A increases. F_B = R_A must increase. Since mu and N_B fixed, max friction fixed. F_B eventually exceeds mu N_B and ladder slips before person reaches A. **M1 A1 A1 [3]**

END OF MARKSCHEME